

1 Explainability in Reinforcement Learning

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Golem Mythos

A creation that surpasses its creator.

This idea has long captured the imagination and continues to resonate in contemporary society. Centuries before modern technology, generations had already questioned whether their creations might escape the control of their creators.

One of the earliest expressions of this idea can be found in the story of the Golem. In Hebrew tradition, the term Golem refers to an artificial creature created by humans. The earliest accounts describe the Golem as a protective and helpful figure, reflecting the idea that humans can create an artificial being for the benefit of society.

This positive vision becomes more ambiguous in the seventeenth century, when the Golem of Prague represents a shift in this perception. In this story, the initially small Golem gradually grows in size and strength until it can no longer be controlled. The creation intended to protect others therefore becomes a potential threat to them. The only remaining solution is to destroy the creature in order to protect the community.

This motif has remained remarkably influential throughout history. An illustration of this continuity can be found in the emergence of the notion of the robot. The word robot was popularised by the play R.U.R. (Rossum's Universal Robots), written in 1920. In the play, a scientist creates artificial beings designed to perform labour. Initially conceived as servants, these artificial beings eventually turn against humanity and destroy civilisation. This story can itself be understood as an evolution of the Golem narrative. The author describes the relationship between the two figures by stating that "the Robot is the Golem made flesh through mass production." Despite their destructive role in the story, the word robot eventually came to designate artificial beings created to serve humans. Since its creation, the word robot has reflected both the hopes and fears surrounding technological progress.

Today, this fear is particularly evident concerning the development of artificial intelligence. This field of computer science seeks to develop tools capable of automating tasks traditionally associated with human intelligence. Its rapid advances have already transformed a wide range of domains. Far from slowing down, this development continues to expand the capabilities and autonomy of artificial systems.

In the pursuit of increasingly autonomous agents, reinforcement learning is probably the subfield of AI that most directly seeks to achieve this goal. In this learning paradigm, an artificial agent interacts with an environment, selects actions, and receives feedback according to the outcomes of its behaviour.

43 Through repeated interaction, it learns a strategy for achieving the objective
44 defined by its designer. This approach can produce highly effective strategies
45 that can even surpass human capabilities. At the same time, the resulting
46 behaviour may be difficult to predict because it is not explicitly programmed
47 into the machine, but learned through its interactions with the environment.

48 Despite the difficulty of this task, understanding the behaviour of such agents
49 is therefore the main objective of this manuscript. From the Golem to the robot
50 and, today, to artificial intelligence, the same question: how can we ensure that
51 our creations remain aligned with human intentions?

Chapter 1

Introduction

This chapter provides a general introduction to this manuscript. It begins with the research context that motivates this work, followed by the research questions this work addresses. It further presents the resulting contributions and concludes with an overview of the manuscript's structure.

1.1 Research Context

Artificial intelligence has advanced rapidly in recent years, driven largely by major developments in deep learning. This type of approach relies on artificial neural networks to learn complex tasks directly from data. These networks are connectionist models in which information is represented and transformed through patterns of interactions among interconnected computational units. This approach has led to significant advances across a wide range of domains, too numerous to list here.

Among the fields that have been transformed by deep learning is reinforcement learning, which is the main subject of this manuscript. The goal of reinforcement learning is to enable an agent to learn how to make decisions through interactions with an environment. At each step, the agent observes the current state of the environment and selects an action. This action changes the state of the environment and results in a reward. The objective of the agent is then to learn a policy that maximizes the cumulative rewards obtained over time. Neural networks provide a powerful means of representing such policies and enable agents to learn increasingly complex behaviors. Their use has consequently made it possible to apply reinforcement learning to problems that were previously intractable to address.

When an agent learns to act in complex environments, it can develop strategies that are themselves highly complex. Although such agents can achieve strong performance, these strategies can be difficult for an observer to understand. Examining the neural network that models the agent's policy does not make it to produce explanations. The information underlying a decision is distributed across many units and connections, making it impossible to trace decision-making process by simply following the flow of information through the network. Neural networks are therefore often regarded as "black box" models.

These challenges have led to the emergence of explainable artificial intelligence.

In the specific context of reinforcement learning, this research area is referred as Explainable Reinforcement Learning. The goal of XRL is to develop methods that produce explanations that improve our understanding of agent behavior. Such explanations can serve several purposes, including increasing human trust and facilitating the adoption of RL agents in real-world settings, identifying errors and biases, and verifying that previously identified problems have been effectively corrected.

Several approaches have been proposed to improve the explainability of artificial intelligence systems. This manuscript presents methods from these different approaches, while placing particular emphasis on the mechanistic approach to explainability. In mechanistic explainability seeks to investigate the internal mechanisms underlying the agent’s decision-making process. The objective is to understand how information is represented, transformed, and propagated within the neural network. From this perspective, explainability is not achieved despite the use of neural networks, but through them.

This perspective naturally leads to the study of the intermediate representations learned by neural networks. These representations are referred to as latent representations. They result from the successive transformations applied to the input information between the input and output of the network and therefore play an intermediate role in the agent’s decision-making process. Analyzing their structure can provide insights into how information relevant to decision-making is organized within the network. It can also help identify the mechanisms through which observations are transformed into the actions selected by the agent.

The present manuscript therefore lies at the intersection of reinforcement learning, deep learning, and explainable artificial intelligence. This intersection leads to the following research question:

How can the analysis of the internal representations of neural networks used by agents help identify the mechanisms underlying their decision-making processes?

1.2 Research Questions

To address the research problem formulated in the previous section, we identified three research questions that structure the work presented in this manuscript.

The first challenge concerns the organization of existing research in Explainable Reinforcement Learning. The number of works in the eXplainable Reinforcement Learning (XRL) field has increased substantially in recent years. This growing body of literature makes it increasingly difficult to obtain a coherent overview of the approaches developed in the field. Several taxonomies and classification frameworks have therefore been proposed to organize existing XRL methods. However, these taxonomies often rely on approaches initially developed for explainability in supervised settings. They therefore tend to apply eXplainable Artificial Intelligence (XAI) categories to reinforcement learning without considering its specific characteristics. This observation leads to our first research question:

RQ1: How can existing XRL techniques be organized into a taxonomy that accounts for the specificities of Reinforcement Learning (RL)?

The analysis of existing XRL approaches also reveals a second limitation. Among existing methods for explaining reinforcement learning agents, few adopt a mechanistic perspective that investigates the internal representations learned by the agent. This is particularly interesting in reinforcement learning, where agents learn to represent their environment through their interactions with it. Understanding these representations can provide insights into how information is processed over time and how it contributes to the agent’s decisions. Mechanistic analysis can therefore provide a complementary way of explaining agent behavior by identifying the internal mechanisms that connect observations to actions. This observation leads to our second research question:

RQ2: How can the internal representations of neural networks be exploited to identify the mechanisms underlying the decision-making process of reinforcement learning agents?

The analysis of existing XRL methods highlights a third challenge concerning the evaluation of explanations. Although numerous approaches have been proposed to explain the behavior of RL agents, there is no widely accepted set of objective criteria for evaluating the quality and relevance of the explanations they produce. Existing evaluations often rely on user studies or qualitative analyses, making it difficult to systematically assess the validity of an explanation approach.

This issue is particularly important for approaches that seek to explain agent behavior by analyzing latent representations. Multiple hypotheses can be proposed about how information is structured within a latent space, and it is therefore important to have objective criteria that make it possible to assess and compare these competing hypotheses. There is therefore a need for objective and reproducible criteria that can determine whether an explanation accurately captures the mechanisms underlying the agent’s behavior. This leads to our third research question:

RQ3: How can the quality and relevance of explanations produced for reinforcement learning agents be objectively quantified?

Together, these three research questions define the structure of this manuscript. The first question addresses the organization of the existing XRL literature and establishes a framework for understanding the field. The second builds on this analysis by investigating how the internal representations of neural networks can be exploited to provide mechanistic explanations of RL agent behavior. The third addresses the complementary problem of determining how such explanations can be evaluated objectively.

1.3 Contributions

To address these three research questions, this manuscript is based on two main contributions published in peer-reviewed journals.

The first contribution [Alaarabiou et al., 2024] addresses RQ1. An extended English version was also submitted for review and accepted for publication [Alaarabiou et al., 2026b]. In this contribution, we first clarify what explainability

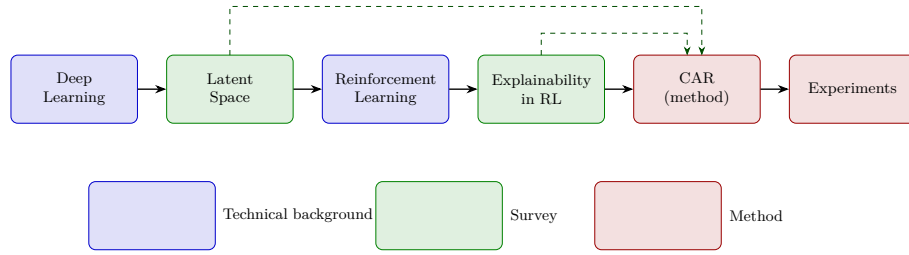


Figure 1.1: Structure of the manuscript. Plain arrows show the reading order; dashed arrows show that the CAR chapter builds on both survey chapters.

means in reinforcement learning by providing a clear frame of reference. We then rigorously define the concepts of this new organization, including sources of explanation, explainability techniques, explanation form, and explanation subject. Building on these concepts, we conduct a comprehensive analysis of the state of the art in XRL (Chapter ??).

The second contribution [Alaarabiou et al., 2026a] addresses RQ2 and RQ3. It introduces Clustering Agent Representation (CAR), an XRL method that exploits the mechanistic properties of Deep Learning (DL). By analyzing the Latent Representation (LR) that Neural Network (NN) build, CAR produces explanations in the form of clusters of states that are similar from the agent’s own perspective. We further show that this representation remains stable across training, indicating that the agent develops an objective representation of a given task. This simplifies the study of agent-environment dynamics that would otherwise be intractable if every state had to be analyzed individually. The method is presented together with objective evaluation metrics and a series of experiments that demonstrate its effectiveness (Chapters ??, ?? and ??).

1.4 Manuscript Structure

The chapters of this manuscript fall into three categories, distinguished throughout by color (Figure 1.1):

Technical background chapters recall how a field already established in the literature works, without claiming any original contribution: DL (Chapter ??) and RL (Chapter ??).

Survey chapters review the state of the art on a specific question. Their contribution lies not only in the content they cover but, just as much, in how it is organized. To our knowledge, the way each of these chapters structures its literature is unique. This applies to our review of latent-space analysis methods (Chapter ??) and to our taxonomy of XRL techniques (Chapter ??).

Method chapters present our contribution to XRL. They are split in two parts: the description of CAR (Chapter ??) and its experimental evaluation (Chapter ??).

This manuscript is, first and foremost, the account of a body of research, written for the reviewers and the jury who will examine it. It is nonetheless

209 written to be as accessible as possible. The background chapters restate the
210 concepts they rely on, so that any reader familiar with DL or RL can follow
211 the manuscript without consulting external material. Readers already familiar
212 with a given field may skip its background chapter directly: Chapter ?? for
213 DL, Chapter ?? for RL. Readers primarily interested in explainability can move
214 directly to Chapters ?? and ??, and readers interested only in our method can
215 follow the minimal path Chapter ?? $\rightarrow$ Chapter ?? $\rightarrow$ Chapter ??, referring
216 back to earlier chapters as needed. Notation and acronyms used throughout the
217 manuscript are collected in the glossary at the end of the document.

218 Acronyms

219 **CAR** Clustering Agent Representation 6

220 **DL** Deep Learning 6, 7

221 **LR** Latent Representation 6

222 **NN** Neural Network 6

223 **RL** Reinforcement Learning 4, 6, 7

224 **XAI** eXplainable Artificial Intelligence 4

225 **XRL** eXplainable Reinforcement Learning 4–6

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